# 2025-03 Patchbox OS on Pi 400

I have a pi 400 that is not being used. Could I test that with Patchbox OS?

Yes! You can absolutely test **Patchbox OS** on your **Raspberry Pi 400** before setting it up on the Pi 4.

### Is the Raspberry Pi 400 Compatible with Patchbox OS?

✅ **Yes**, the Pi 400 works because it's based on the **Raspberry Pi 4 architecture** (same CPU/GPU).
✅ It **has better cooling** than the Pi 4, so it runs smoothly.
✅ **Works with USB audio interfaces & HDMI audio extractors** just like the Pi 4.
❌ **No built-in GPIO header** – so you **can't** directly connect your HiFiBerry DAC without an adapter.

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### How to Install Patchbox OS on Your Pi 400

**1. Download & Flash the OS**

- Download **Patchbox OS** from Blokas.io
- Flash it to a microSD card using **Raspberry Pi Imager** or **Balena Etcher**
- Insert the microSD card into your Pi 400 and boot it up

**2. Initial Configuration**

- When prompted, run:

```bash
patchbox-config
```

- Select **"Audio Setup"** and choose your audio output:
  - **HDMI Audio** → If using a monitor with built-in speakers
  - **USB DAC** → If using a USB sound card (better quality)
  - **3.5mm Jack from HDMI Adapter** → If using an HDMI audio extractor

**3. Test MIDI & Synth Playback**

- Install **FluidSynth** to test piano sounds:

```bash
sudo apt install fluidsynth
```

- Download a good piano SoundFont:

```bash
wget https://www.schristiancollins.com/soundfonts/FluidR3_GM.sf2 -O ~/piano.sf2
```

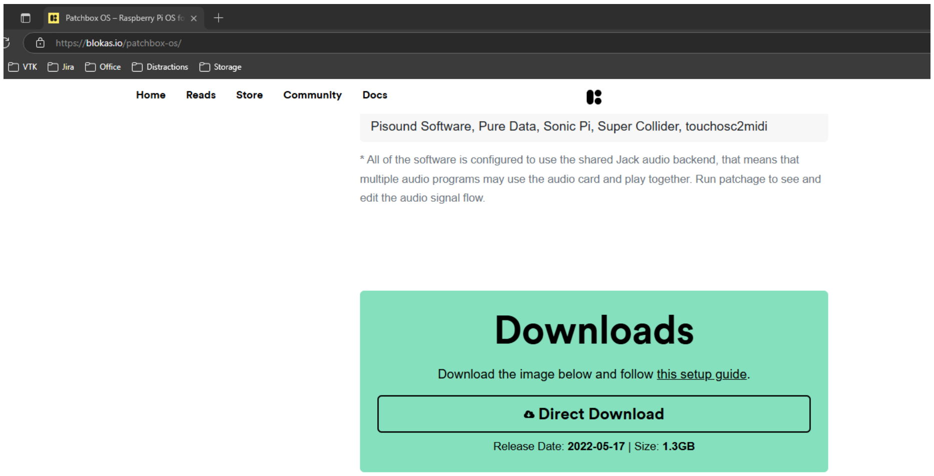
- Play a test note:

```bash
fluidsynth -a alsa -m alsa_seq ~/piano.sf2
```

**4. Connect MIDI Keyboard (Optional)**

- If your dad's piano has a **USB MIDI Out**, plug it into the Pi 400.
- If it has a **5-pin MIDI Out**, you'll need a **USB MIDI interface** (like the Roland UM-One or a cheap MIDI-to-USB adapter).

Patchbox OS – Raspberry Pi OS for Audio Projects




Install the Operating System to the SD Card - Patchbox OS Documentation

balenaEtcher - Flash OS images to SD cards & USB drives

## Step 01: Install balenaEtcher

### Windows

1. Download balenaEtcher
2. Double-click the *.exe file and follow the setup wizard
3. Run balenaEtcher in the Administrator Mode: right-click on balenaEtcher and choose ‘Run as administrator’

## Step 02: Download the OS Image

1. Download a copy of the OS image you want to install (the Patchbox OS image can be downloaded here)
2. Unzip the file after it has downloaded

## Step 03: Flash the SD Card

1. Insert the SD card to your computer's SD card reader
2. Launch balenaEtcher
3. Click **Flash from file** and select the file you want to upload (e.g. `2020-03-14-Patchbox.im`
4. Click **Select target** and choose your SD card
5. Now click **Flash!** to write the image file to the SD card
6. When done, safely remove the SD card, insert it into your Raspberry Pi and power it on.

**Note:** If you want to flash another SD card with the same image, insert it and click **Flash Another**.